

Week 40, Tuesday Session

Flipped classroom: your questions, then Project 1 only – what your results say, what you have learned, and how to present it

Bring your laptop, your figures and your draft

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How today works

The format (as every Tuesday)

Mondays are lectures; Tuesdays belong to your questions and to concrete cases. Today the cases are *your own results*: the project you hand in on Monday.

1. **First ~20 min:** your open questions from Monday's lecture – the neuron and its two sweeps, XOR and the hidden layer, the universal approximation theorem, activation functions, the scalar backpropagation examples and `jax.grad` – discussed in plenum.
2. **Rest of the double session:** *two* cases, worked in your project groups with your Project 1 code, figures and draft open. Case 1: model selection and the bias–variance story (parts a–d and i) – what do your figures say, and do they agree with each other? Case 2: the optimisers and Lasso (parts e–h) – what did you actually learn, and how do you compare methods honestly? Each case ends with the questions a grader – or an examiner – will ask you.
3. **Last 5 min:** wrap-up, how to hand in, and what Monday builds on.

No weekly exercises this week

The week is for finishing the project. **Project 1 is due Monday October 5.**

Picking up from Monday – and where you should be with Project 1

Monday in one sentence each

The neuron: logistic regression with a sigmoid; its learning rule is error times the input on the edge. Read as a graph, one reverse sweep gives the whole gradient for two evaluations of the cost; forward mode needs p sweeps. XOR: one neuron draws one line; two hidden nodes learn the two features that make the problem linear. Universality needs a non-polynomial activation and says nothing about width or learnability. Only f' enters the backward pass. Backpropagation on one node per layer: δ is the adjoint of the pre-activation, $\delta^1 = \delta^2 w^2 \sigma'(z^1)$, and `jax.grad` reproduces every digit.

Questions we expect

Why is δ both the bias gradient and the adjoint of z ? Why did the $\gamma = 1$ run stall at 5.02 and never recover? If the theorem guarantees a one-layer network, why go deep? Why is the sigmoid last in Table 8.2 when it is the one Cybenko proved the theorem for? Where does the sum over k come from next week? Ask.

Project 1: what should be done by now

Part	Done in	Status
a) OLS, b) Ridge	wk 35–36	figures final
c) bias–variance, bootstrap	wk 36	derivation done
d) k -fold CV	wk 36	compared with c)
e) GD + AD, f) momentum etc.	wk 37–38	tables
g) Lasso, h) SGD	wk 38–39	tables, figures
i) final CV selection	now	Case 1 today
abstract, intro, conclusion	wk 39	drafted

What remains

The *analysis* – 40 of 100 points – and the text around it; a clean re-run of the whole pipeline from a fresh kernel so that the numbers in the report are the numbers the code produces; the LLM appendix; the GitHub link. Today is about the analysis.

The analysis is 40 points: what a figure must do

“Analysis of results” (EvaluationForm.md)

Are the results presented clearly and *analysed*? Is the analysis tied to the theory? Are the numerical results reliable (uncertainties, dependence on n , σ , seeds)? Are the methods *compared*, and are the comparisons fair? Does the text say what was learned about the method, not only what the numbers are?

Three levels of a result

Reporting: “the test MSE is 0.0121 at degree 7.”

Interpreting: “... where the bias² (0.003) and the variance (0.008) are of the same size, Eq. (2.52).” **Learning:** “... so on this problem the variance, not the bias, limits the polynomial model, which is why Ridge (part b) extends the usable degree and Lasso (part g) does not help.” The 40 points are for the third level; it requires the first two.

A figure should carry one claim

- ▶ Axes labelled with quantity *and* convention (MSE on test data, λ on a log axis, degree as integers).
- ▶ Log scale when the quantity spans decades (MSE, excess cost, λ , γ).
- ▶ Uncertainty visible: bootstrap or CV spread as bands or bars, the seed stated.
- ▶ One message per panel; the caption *states the finding*, not the content: “Test MSE is minimal at degree 7; beyond degree 9 the variance grows faster than the bias falls” – not “MSE vs degree”.
- ▶ Curves that are compared share axes and colours across figures (OLS blue, Ridge red, Lasso yellow – everywhere).
- ▶ Tables: the same number of digits as the uncertainty warrants; units; the best entry marked.

Case 1: model selection and the bias–variance story (parts a–d and i)

Goal: leave with the figures and the one table that carry the model-selection argument of your report, each with a caption that states a finding, and with the paragraph of part i) drafted.

1. **Assemble the evidence** (your own code): (A) train and test MSE against degree for OLS, part a); (B) test MSE against degree *and* λ for Ridge, part b), as a heat map or a family of curves; (C) bias^2 , variance and σ^2 against degree from the bootstrap, part c); (D) CV MSE against degree for $k = 5$ and $k = 10$ on the same axes as the bootstrap MSE, part d).
2. **Write the caption of each** as one sentence that states what the figure shows – with the number in it. Then the “so what” sentence that follows it in the text.
3. **Cross-examine the four:** do bootstrap and CV point to the same degree? Does the MSE minimum sit where $\text{bias}^2 \approx \text{variance}$? Does Ridge move the minimum to higher degree, and by how much? Does your bias^2 curve go to zero at high degree – and if not, why (true f or y)? Change one thing – double n , or halve σ – and say in one sentence what moved.
4. **Part i), the table:** OLS, Ridge and Lasso, each at its cross-validated best (degree, λ), with the CV MSE and its standard error. Draft the paragraph: which model, why, and which term of Eq. (2.52) each method attacks – supported by (C).

Case 1: what you should find – and what it looks like when it is wrong

The picture that should emerge (qualitatively)

(A) Training MSE falls monotonically with degree (Eq. 3.91); test MSE has a broad minimum, then rises – Hastie's Figure 2.11, noisier. (C) Bias² falls steeply up to the degree where the polynomial can follow Runge's peak and then flattens; variance rises with degree, faster once p approaches n_{train} ; the sum tracks the test MSE of (A). (B) Ridge: the minimum moves to higher degree and becomes shallower as λ grows; the best λ falls with n (Eq. 3.95). (D) Bootstrap and k -fold CV agree on the *region*, rarely on the exact degree; $k = 5$ is slightly more pessimistic than $k = 10$; the CV curve is smoother than the bootstrap one for the same number of resamples.

Symptoms, and what they mean

- ▶ Bias² never below σ^2 : you used y , not f , in the bias – say so, or use f (the project allows it).
- ▶ Test MSE below σ^2 : you evaluated on training data, or scaled with the full data before splitting (Section 3.13).
- ▶ Variance decreasing with degree: the bootstrap resamples the test set instead of the training set.
- ▶ CV and bootstrap differ by a factor two in level: different training-set sizes (63% unique points in a bootstrap sample against 80–90% in a fold) – a known effect, worth a sentence.
- ▶ Ridge MSE independent of λ below 10^{-6} and equal to OLS: fine, that is what $\lambda \ll \sigma_{\min}^2$ means, Eq. (3.48); show the axis where it starts to matter.
- ▶ Different seed, different best degree: the minimum is flat – report the range, or the one-standard-error rule (Section 3.14).

Case 1: do you have the insight?

1. Your bootstrap analysis puts the test-MSE minimum at degree 8, your 10-fold cross-validation at degree 6. Is one of them wrong? What do you report as the best model, and what number do you put in the abstract?
2. Your bias^2 curve flattens at 0.011 at high degree while $\sigma^2 = 0.01$. Explain the plateau, say what it would look like with the true f , and decide which version belongs in the report.
3. Ridge with $\lambda = 10^{-3}$ extends the usable degree from 8 to 12. Which term of Eq. (2.52) did it reduce, by what mechanism (Section 3.8), and what did it cost? What single plot would show both?
4. You re-ran the pipeline with a new seed and the OLS minimum moved from degree 7 to 9, the MSE at the minimum from 0.0121 to 0.0118. Is your conclusion invalidated? What *is* the reproducible result here?

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What we hope you say

(1) Neither. Both estimate the same expected test error with different resampling schemes; the minimum is flat over degrees 6–9, and inside a flat region two noisy estimators pick different points. Report the region and the MSE level with its standard error, choose the *simplest* model within one standard error of the minimum (Section 3.14), and put the level, not the degree, in the abstract. (2) Replacing f by y in the bias adds σ^2 to it (the noise is not fitted by the mean model), so the plateau is σ^2 plus the true bias, here 0.001. With f the curve goes to ≈ 0.001 and the three terms sum to the test MSE exactly. Report the version with f , since f is known here, and state that on real data the y version is all one can compute. (3) The variance: Ridge multiplies mode i of the fit by $\sigma_i^2/(\sigma_i^2 + \lambda)$, Eq. (3.48), suppressing the directions with tiny singular values that the high-degree columns produce; the price is bias, visible as a systematic underestimate of the peak of f . One plot: bias² and variance against degree for OLS *and* Ridge on the same axes. (4) No. The degree of the minimum is a noisy statistic of a flat curve; the reproducible result is the MSE level (0.012 ± 0.001), the flat region, and the mechanism – and the seed goes in the report so that the figure can be regenerated.

Case 2: the optimisers and Lasso (parts e–h): is the comparison fair?

Goal: one table and two figures that compare the methods on equal terms, a clear statement of what your Lasso code does at zero, and the critical paragraph of your conclusion drafted.

1. **The table (parts e–f):** for one degree (say 5) and one Ridge λ , every method – plain, momentum, AdaGrad, RMSProp, Adam – with its best γ , the number of iterations to reach $\|\theta - \hat{\theta}_{\text{closed}}\| < 10^{-6}$ (or a stated tolerance), and the final distance. Add λ_{max} of the Hessian $\frac{2}{n}X^T X$, $\gamma_{\text{max}} = 2/\lambda_{\text{max}}$ and κ (Eq. 4.20, 4.26) – with and without scaling. Add the one number from part e): $\max|\nabla C_{\text{analytic}} - \nabla C_{\text{AD}}|$.
2. **The γ scan (part f):** iterations to tolerance against γ on a log–log axis, one curve per method: the *usable window* of each, and what happens at its edges.
3. **SGD (part h):** excess MSE against *epochs* for $M = 1, 5, 25$ and full batch, at a constant and at a decaying γ_t ; state the flops per epoch (Eq. 4.38) so that the comparison is per pass over the data, not per update.
4. **Lasso (part g):** what does your AD return for $d|\theta|/d\theta$ at 0? Does your code produce *exact* zeros – and which coefficients should be zero for Runge’s function on a symmetric grid? Compare with `scikit-learn` after matching the conventions $(\alpha, 1/(2n))$.
5. **The critical paragraph:** which method you would use, at what cost, what the comparison could not resolve, and what you would do differently.

Case 2: what you should find

Gradient descent and its variants (weeks 37–38)

$\lambda_{\max}$ is of order one but κ grows fast with degree because the columns x^k are collinear (week 37: degree 5, $\kappa = 520$; degree 10, $\sim 10^6$); standardising removes scale differences, not the collinearity. Plain gradient descent converges only for $\gamma < 2/\lambda_{\max}$ and needs $\sim \kappa \ln(1/\epsilon)$ iterations (Eq. 4.26); momentum buys $\sim \sqrt{\kappa}$ (Eq. 4.33) and its usable window extends to $2(1 + \beta)/\lambda_{\max}$; AdaGrad's step decays as $1/\sqrt{t}$ and it stalls unless γ is large; RMSProp at a constant γ rattles at a floor $\propto \gamma$; Adam is insensitive to γ over a decade and rarely the fastest to 10^{-8} (week 38: Runge degree 6, plain 7346, momentum 670, Adam 1062 iterations). Ridge lowers κ and every method speeds up. The AD and analytic gradients agree to 10^{-15} – if not, a convention ($1/n$, factor 2, λ) differs between the two.

SGD (weeks 38–39)

Per *update* SGD is noisy; per *epoch* it is far ahead of gradient descent early on, and at a constant γ it equilibrates in a cloud $\propto \gamma/M$ (Eq. 4.50); a decaying schedule reaches the closed-form solution, a bad one (t_0, t_1 too small) freezes. With $M = 1$ the stability bound holds *per point* – unscaled features diverge at a γ that was safe for the full batch.

Lasso (weeks 35–36, 39)

$\partial|\theta|/\partial\theta$ at 0 is undefined; the subgradient is $[-1, 1]$; JAX returns 1.0 (a valid element, not a special one). Plain subgradient descent does *not* produce exact zeros – coefficients hover within $\sim \gamma\lambda$ of zero. For Runge's function on a symmetric grid the odd coefficients should vanish (the function is even): a check that the penalty *selects*, not merely shrinks. `scikit-learn` minimises $\frac{1}{2n}\|y - X\theta\|^2 + \alpha\|\theta\|_1$ by coordinate descent with soft-thresholding, Eq. (3.63) – exact zeros, and $\alpha = \lambda$ only if your cost carries the same $1/(2n)$.

Case 2: do you have the insight?

1. With standardised features, plain gradient descent at $\gamma = 0.4$ converged for degree 3 and diverged for degree 8 on the same data. Why – and how do you set γ for every degree without trial and error?
2. Adam reached the closed-form solution to 10^{-4} for every γ between 10^{-3} and 1, but never below 10^{-6} at a constant γ , while tuned plain gradient descent did. Which method “wins”, and what must a fair comparison specify?
3. Your minibatch run with $M = 1$ diverged at a γ that was safe for the full gradient. Explain, and give two fixes without guessing.
4. Your Lasso code gives $\theta_3 = 2 \times 10^{-4}$ where `scikit-learn` gives exactly 0, and its bootstrap standard deviation is 10^{-2} . Are these different results? What do you write?

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What we hope you say

(1) $\lambda_{\max}(\frac{2}{n}X^TX)$ grows with the degree (more columns, and nearly collinear ones: on 100 standardised points 3.8 at degree 3, 7.6 at degree 8), so the bound $\gamma < 2/\lambda_{\max}$ (Eq. 4.20) falls from 0.52 to 0.26 – past 0.4. Compute $\lambda_{\max}$ once per design matrix (one `np.linalg.eigvalsh`) and set γ to a fixed fraction of $2/\lambda_{\max}$; mind the convention, a cost without the $1/n$ has $\lambda_{\max}$ n times larger. Standardising removes differences of *scale*; the collinearity of x^k on $[-1, 1]$ it cannot remove, so κ stays large at high degree and Ridge helps the optimiser too. (2) Neither, as stated. A comparison needs a *target accuracy* and a *cost*: to 10^{-4} Adam is the robust choice (no tuning); to 10^{-8} plain or momentum with a tuned γ , or Adam with a decaying γ . Report iterations to a stated tolerance and the γ window over which each method reaches it – “best” is a function of the question. (3) The stability bound holds per minibatch: for $M = 1$ the relevant Hessian is $2x_i x_i^T$ with $\lambda_{\max} = 2|x_i|^2$, up to 86 for degree 5 on $[-1, 1]$ (week 38), far above the full-batch $\lambda_{\max}$. Fixes: set $\gamma < 2/\max_i 2|x_i|^2$ (one line of code), or take M large enough that the minibatch Hessian is close to the full one – computed, not guessed. (4) The same result. Subgradient descent hovers within $\sim \gamma\lambda$ of zero and never lands on it; coordinate descent with soft-thresholding does. A coefficient of 2×10^{-4} with a standard deviation of 10^{-2} is zero within its uncertainty, and the odd power *should* be zero by the symmetry of Runge’s function. Write: “our gradient-based Lasso drives the odd coefficients to $|\theta_k| < 10^{-3}$, zero within the bootstrap uncertainty; exact zeros require the soft-threshold step of Eq. (3.63), which `scikit-learn` uses” – after matching the α convention.

The last checks before Monday

Reproducibility

- ▶ Restart the kernel, run everything top to bottom, and compare every number in the report with the output. Seeds fixed and stated.
- ▶ Every figure saved as PDF by the code (`plt.savefig`), named after what it shows, and produced by a script in the repository – a reader must be able to regenerate it.
- ▶ The closed-form OLS and Ridge solutions are the *tests* of your gradient code: quote the agreement.
- ▶ You can upload your Jupyter notebook to the GitHub repository – executed top to bottom, so that a reader sees the outputs.

Hand-in (Canvas, Monday October 5)

- ▶ One PDF, the report; the code as a link to your GitHub/GitLab repository with a folder of selected results and a README that says how to run it.
- ▶ The LLM appendix after the conclusions – missing means –5 points – consistent with the LLM-assisted docstrings.
- ▶ References verified against the sources, in the course format.
- ▶ Group of two to three: all names on the report, one hand-in.

Wrap-up, the deadline, and Monday

- ▶ **No weekly exercises. Project 1 is due Monday October 5** – PDF on Canvas, code on GitHub, LLM appendix included.
- ▶ **Next Monday (week 41):** the general backpropagation equations with the sum over receiving nodes (Sections 8.8.3–8.8.5), vanishing gradients and initialisation, and how to write your own neural-network code (Section 8.13) – the core of Project 2, which applies it to the regression problem you just finished and to a classification problem.